CS-433 Machine Learning 2025/26 - Project 1

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I. Introduction

Cardiovascular diseases (CVD) are the leading cause of death globally, emphasizing the need for accurate and interpretable risk prediction models [1]. Machine learning offers strong predictive capabilities, but limited transparency often restricts clinical adoption.

This study aims to explore three separate models and compare their predictive capabilities as well as their transparency and explainability to identify the model best suited for explainable cardiovascular risk prediction.

II. Data overview

The BRFSS dataset is an ongoing surveillance system designed to measure behavioural risk factors among the healthy adult population residing in the United States [2].

The full dataset contains 441,456 samples with 321 features. In this study, an 80/20 split provided by the course was used for training and testing. The MICHD outcome is negatively skewed by a factor of more than 10, meaning that most survey participants did not have MICHD. The data include information on participants' health-related behaviours (such as alcohol intake and exercise), chronic health conditions (cardiovascular and other diseases), and additional lifestyle factors [2].

III. Models and Methods

A. Model selection

The goal of this study is to develop a model suitable for use by medical experts in the high-risk field of cardiovascular disease. Therefore, the model's decisions must be clear and explainable to instill confidence and enable intervention when necessary. Motivated by these goals, three models were selected: the ID3 decision tree algorithm, K-nearest neighbours (KNN), and a neural network.

B. Performance metric selection

The F1 score was used for both model evaluation and hyperparameter tuning. This metric was selected because of its ability to balance precision and recall equally. This balance is important because, as previously mentioned, the dataset is heavily skewed. In addition, MICHD is a life-altering condition; therefore, compared to the cost of further examination, false positives should not be heavily penalized.

C. Data Preprocessing

As the BRFSS dataset is based on a telephone survey, many of its features contain blank or non-displayed data (see “IDATE” in [3]). Since explainability of lifestyle factors affecting cardiovascular outcomes was a high priority, it was crucial to ensure that only a relevant subset of features was used.

To address missing data, all columns with more than 30% missing values were dropped, as interpolating these could compromise model integrity. Columns with 5–30% missing values were selectively removed based on the following criteria: correlation with the target variable above a p-value of 0.05 and collinearity greater than 80%. Initially the remaining columns were interpolated using the median due to the presence of many outliers. However, hyperparameter tuning showed that mean interpolation yielded better performance.

After data cleaning, preprocessing included the following steps: removal of zero-variance columns, standardization of features, and clipping of outliers beyond 5σ. Finally, due to the large class imbalance, the training dataset was resampled at a 2:1 negative-to-positive ratio, which was selected through hyperparameter tuning.

D. ID3 Decision Tree

A decision tree is an arrangement of tests that prescribes an appropriate test at each step of the analysis [4]. Once the tree is built, every step of the dataset analysis is clearly defined. Given a sample, generating a label consists of following a path from the root node to a leaf of the tree, which corresponds to the label to be assigned.

The ID3 algorithm builds such a tree by maximizing information gain at each step (or tree node), that is, by minimizing the entropy of the selected attribute. The dataset is then split according to this attribute to produce subsets of the data, on which the algorithm recurs. Because of its tree-like structure, the model is easily interpretable.

The values of each feature were divided into a constant number of categories or ”bins”, such that the categories uniformly partitioned the value range. This approach was necessary because the model is only feasible when the number of possible feature values is limited [4].

The ID3 decision tree then has essentially two hyperparameters to control: the bin size for numerical data and the maximum tree depth. Larger bins result in less precision, which can lead to more false positives. However, as false positives are not heavily penalised, a rough estimate is acceptable, and a bin size of 3 was selected. A tree with larger depth can improve predictive performance but increases the risk of overfitting to the training data. To prevent overfitting, an appropriate tree depth was selected by optimising the mean F1-score over folds in k-fold cross-validation.

Due to hardware limitations and (in)efficiency of ID3, the number of features is an important consideration. To address this, principal component analysis (PCA) was applied to reduce dimensionality, retaining only the components responsible for the greatest variance under the assumption of linearity. The reduction to 50 dimensions was determined through hyperparameter tuning.

Because ID3 proved too computationally demanding on the full dataset of over 300,000 samples, subsampling was applied to obtain a manageable subset on which the model could be executed efficiently.

E. K-nearest neighbours classification

K-nearest neighbours excels in binary classification [5]. Supporting this study's goals, KNN is highly transparent and in a real deployment. Its decisions can be readily inspected and explained to medical professionals and patients. KNN has weaker global interpretability compared to ID3; however, through permutation importance, trends can still be extracted from the model to understand feature influence [5].

Similar to ID3, KNN performs better with lower dimensionality. Using PCA, the dataset was reduced to 50 dimensions, a value selected through hyperparameter tuning. The initial implementation computed the full pairwise distance matrix, which required allocating a 1.59 TB array. To overcome this memory limitation, the test and training sets were divided into manageable batches, and distances were computed incrementally between each test batch and subsets of the training data. For each test instance, the algorithm maintains only its current top-k nearest neighbors, updating this list as new training blocks are processed. This avoids allocating the massive $N_{test} \times N_{train}$ distance matrix while still producing exact KNN results. The number of nearest neighbours was chosen to be 16 through hyper parameter tuning.

F. Using neural networks

Neural networks are a widely used class of models in machine learning. In this task, a simple feedforward neural network was implemented, using ReLU functions as activation functions and a sigmoid function in the final layer to map outputs to the (0, 1) range. The genetic algorithm was chosen for training. Data preprocessing was applied as discussed earlier. Multiprocessing was employed to accelerate training, as neural networks in the genetic algorithm's population can be evaluated in parallel during a single iteration.

An F1 score of 0.36 was achieved. During experimentation, several hyperparameters were varied, including the size of the neural network (both the number of layers and the size of each layer), the number of dimensions after dimensionality reduction, the number of iterations, and the number of samples used from the dataset. None of these adjustments had a significant effect on the F1 score, which may indicate a fundamental limitation of either the basic neural network architecture or the optimization algorithm used for training.

IV. Results

The results were obtained using the preprocessing pipeline described earlier and the provided test dataset. Table I summarizes the performance of each model.

Table I
Model Performance Comparison

Model	PCA Dimensions	Balance Ratio	F1 Score
ID3	50	2:1	0.221
KNN	50	2:1	0.373
NN	35	2:1	0.360

Among the evaluated models, K-nearest neighbours achieved the highest F1 score of 0.373, followed closely by the neural network with 0.360. The ID3 decision tree performed noticeably lower, with an F1 score of 0.221.

V. Discussion

From Table I, it can be observed that KNN achieved the best predictive performance while offering a relatively high level of transparency. The neural network performed comparably but lacked the same level of interpretability, which may limit its applicability in high-stakes medical contexts where decision explanations to patients are essential.

The ID3 decision tree produced the lowest F1 score but remains the most globally explainable model. Its structure provides clear insight into how predictions are made, which can be valuable for understanding key risk factors and communicating results to patients.

Given these characteristics, a hybrid approach is suggested. Using KNN for prediction due to its higher accuracy, and ID3 for post-hoc analysis or validation where explainability is critical.

VI. Conclusion

This study developed three machine learning models to predict the risk of developing coronary heart disease (MICHD) using non-strictly-medical data collected from a survey. After evaluating the performance and explainability of an ID3 decision tree (F1 score 0.221), a K-nearest neighbours model (F1 score 0.373), and a neural network (F1 score 0.360), the K-nearest neighbours model was selected. The results indicate that this model can effectively predict MICHD risk and may assist medical professionals in the early detection and prevention of the disease.

References

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